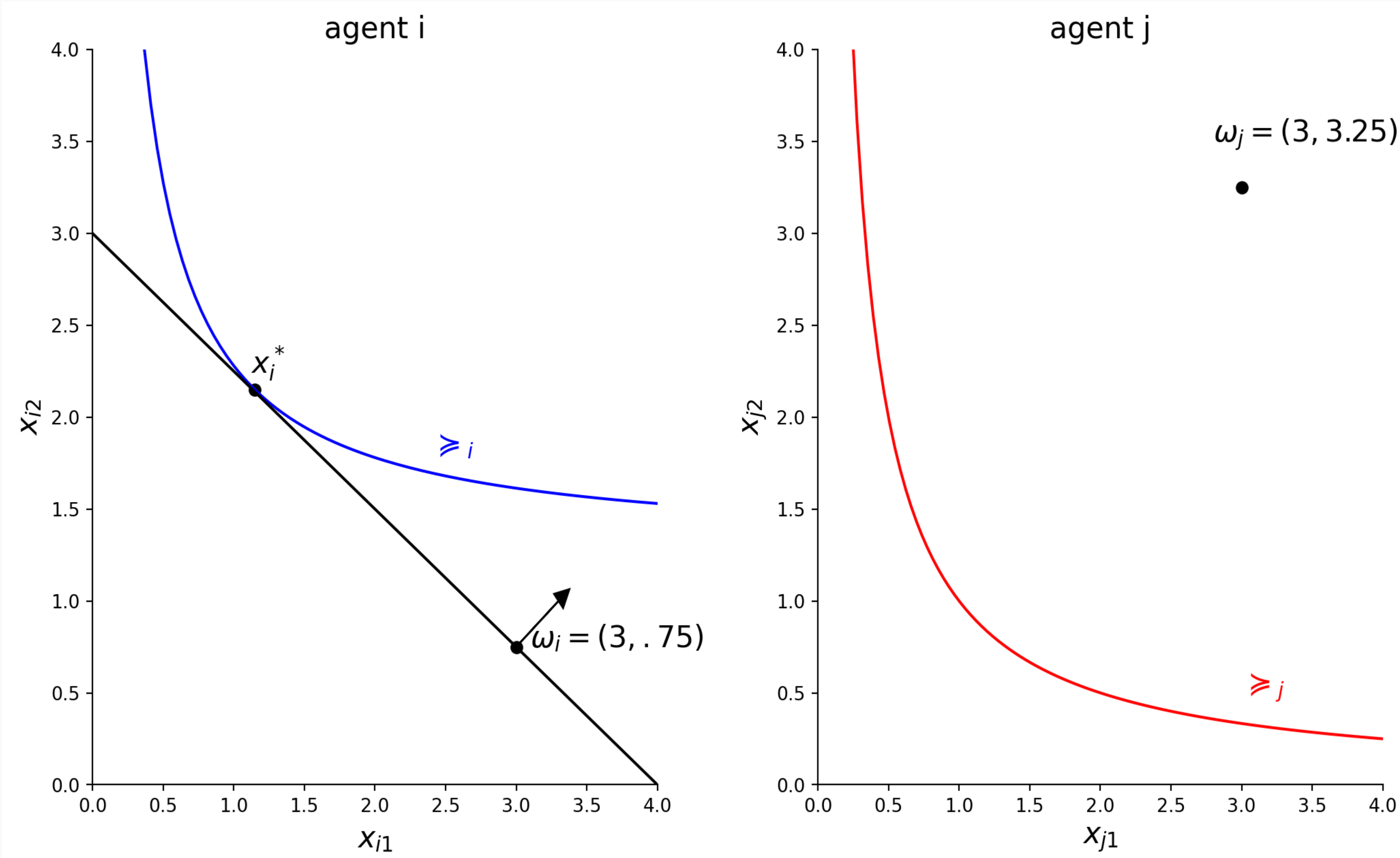
Basic graphs

Edgeworth box

- It is a graphic representation of an exchange economy; no production.
  - 2 goods: $L = 2$, $\ell = 1, 2$; commodity space is $\mathbb{R}^2$.
  - 2 consumers: $I = 2$, $i = 1, 2$.
  - Consumption sets $X_i = \mathbb{R}^2_+$,
  - Preferences $\succeq_i$ complete, reflexive, transitive, continuous, convex, strongly monotone.
  - $\omega_i = (\omega_{i1}, \omega_{i2}) \in \mathbb{R}^2_+$, $\omega_{i\ell}$ is i's endowment of good ℓ.
- Aggregate endowment $\bar{\omega} = \omega_1 + \omega_2$.
- Allocation $(x_1, x_2) \in \mathbb{R}^4_+$ to graph in 2 dimensions.

Graphing an Edgeworth box



Elements of an Edgeworth box

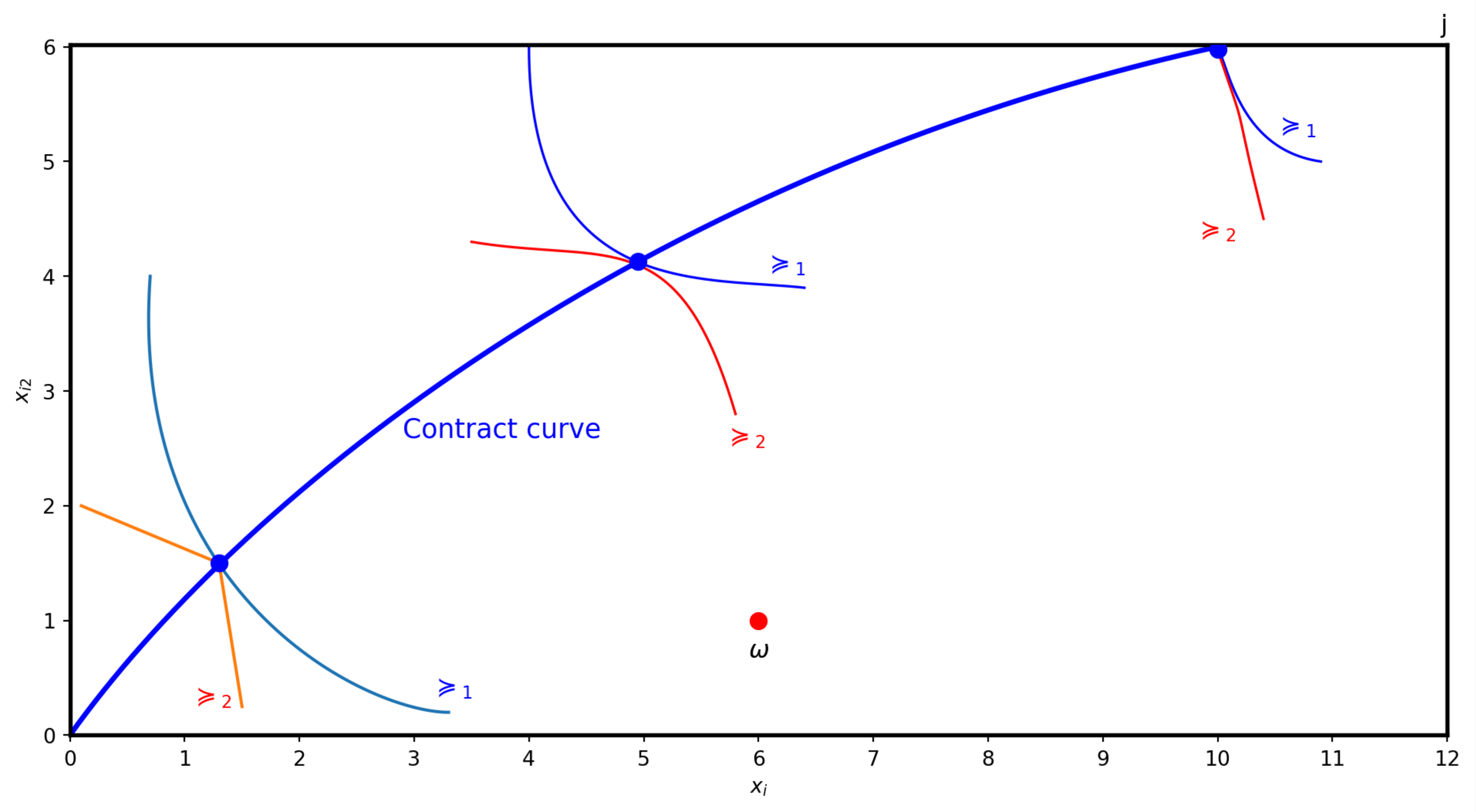
Edgeworth box

Allocations

- A (feasible) allocation is an element x in $(\mathbb{R}^2_+)^2$ with $x_1 + x_2 = \bar{\omega}$.
  - It is sometimes called an *exact allocation* as opposed to requiring $x_1 + x_2 \leq \bar{\omega}$.
  - Every exact allocation is a point in the Edgeworth Box and vice versa.

- A price $p \in \mathbb{R}^2_+$ is represented as a hyperplane in $\mathbb{R}^2$, a straight line.
- Budget set: $B_i(p, \omega_i) = \{x \in \mathbb{R}^2_+ : p \cdot x \leq p \cdot \omega_i\}$. Note that $B_i(p, \omega_i)$ may extend outside the Edgeworth box.

Pareto optimality-Planner's problem



Graph explanation

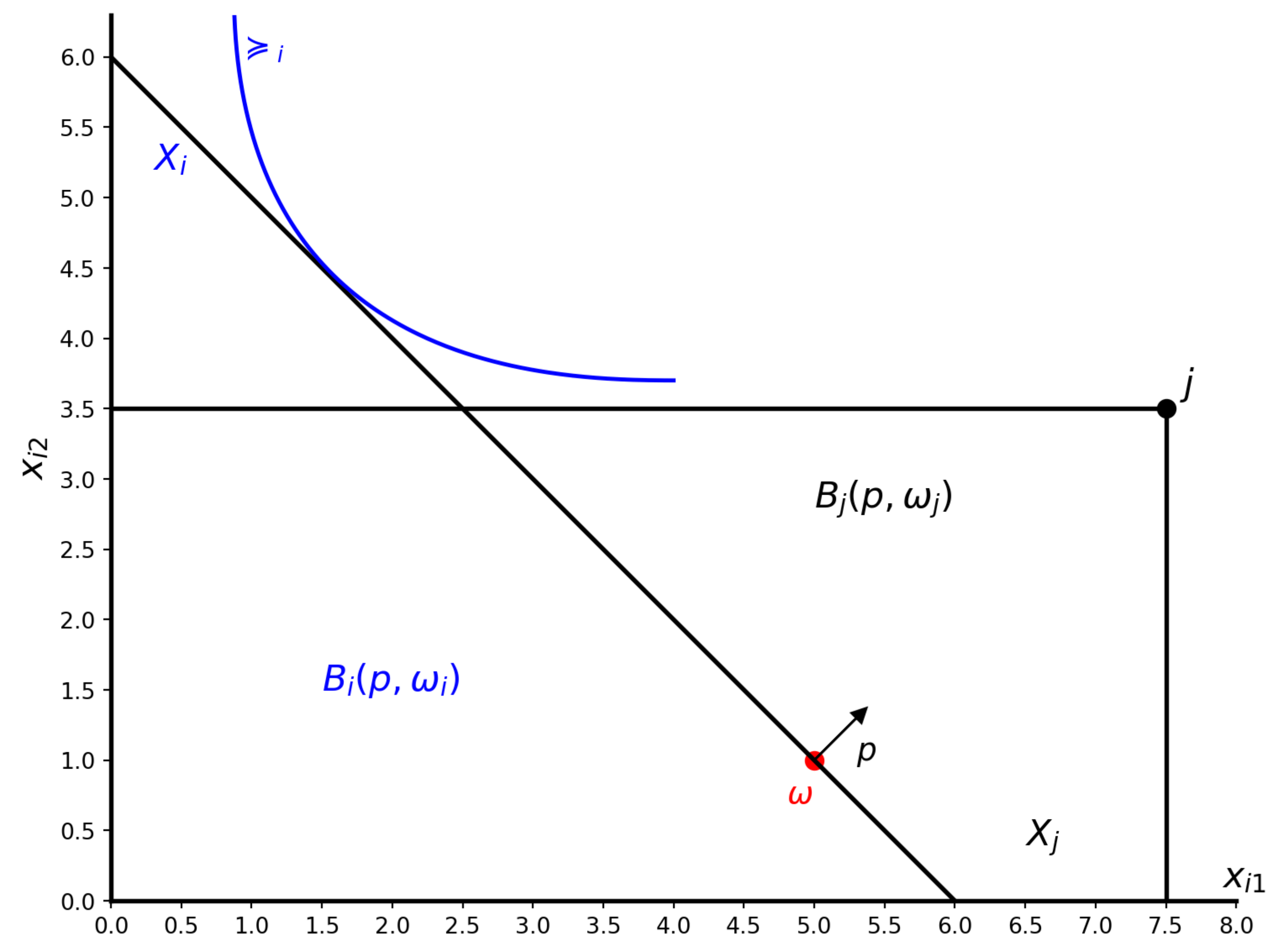
The previous graph illustrates the following points.

- Interior optimal allocation without differentiable indifference curves, a kink Pareto optima.
- Interior Pareto optimal allocation with differentiable preferences.
- Corner Pareto optimal allocation; the allocation is at the boundary of player j's consumption set. This may occur with or without differentiable preferences.
- With differentialbe preferences, first order conditions characterize Pareto optimality at an interior point, see Mas-Colell, Whinston, and Green (1995), 16.F, page 561.

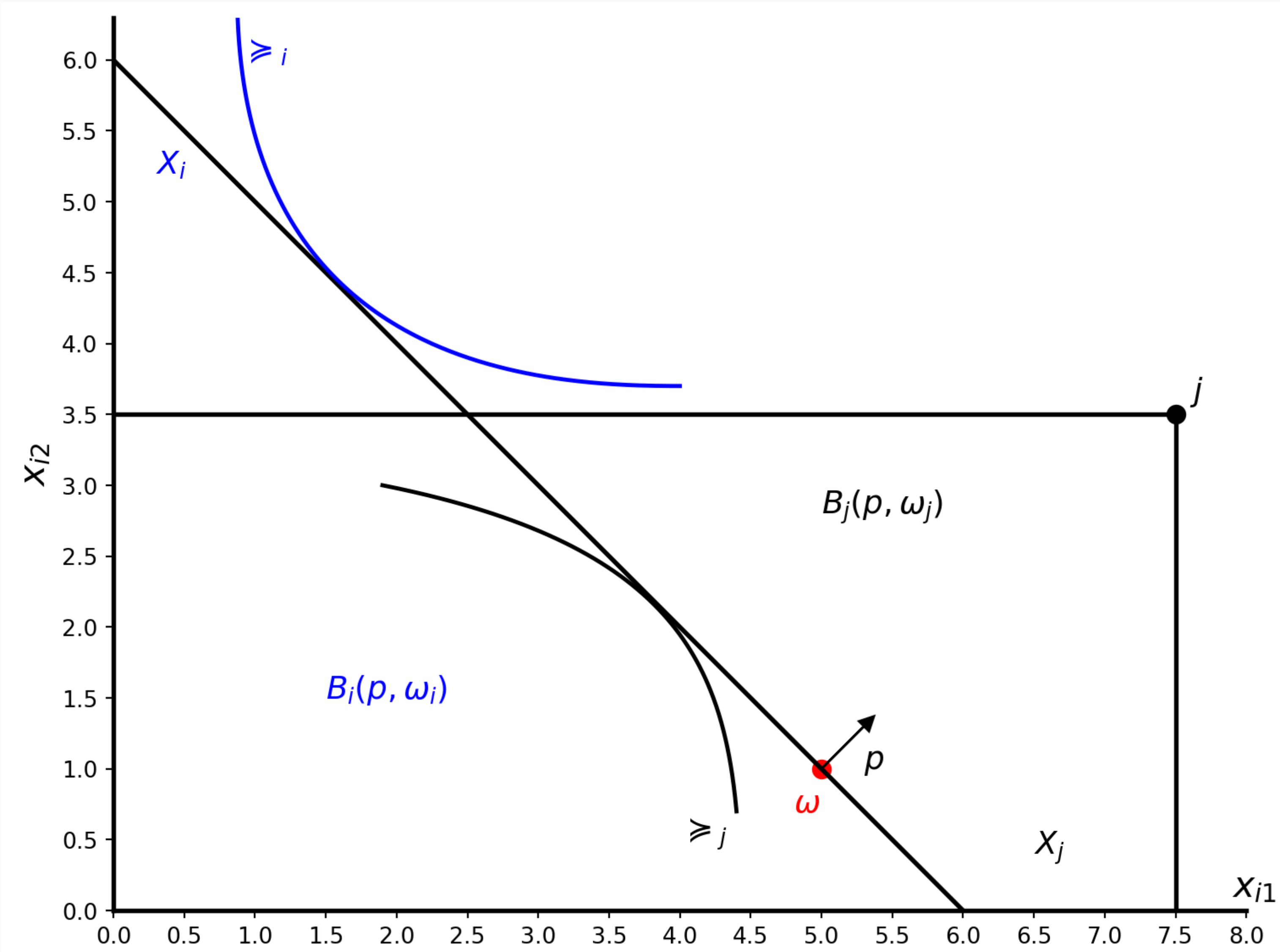
Pareto optimality in the Edgeworth box

- If preferences are smooth, interior Pareto optimal allocations may be computed by equating Marginal Rates of Substitutions $MRS_1 = MRS_2$.
- If a PO allocation lies on the boundary of the Edgeworth box, tangency may fail, i.e. $MRS_1 \neq MRS_2$. This is common as agents typically consume a small fraction of available goods.
- If preferences are not smooth, Pareto optima allocations need not be points of tangency, i.e. $MRS_1 \neq MRS_2$.

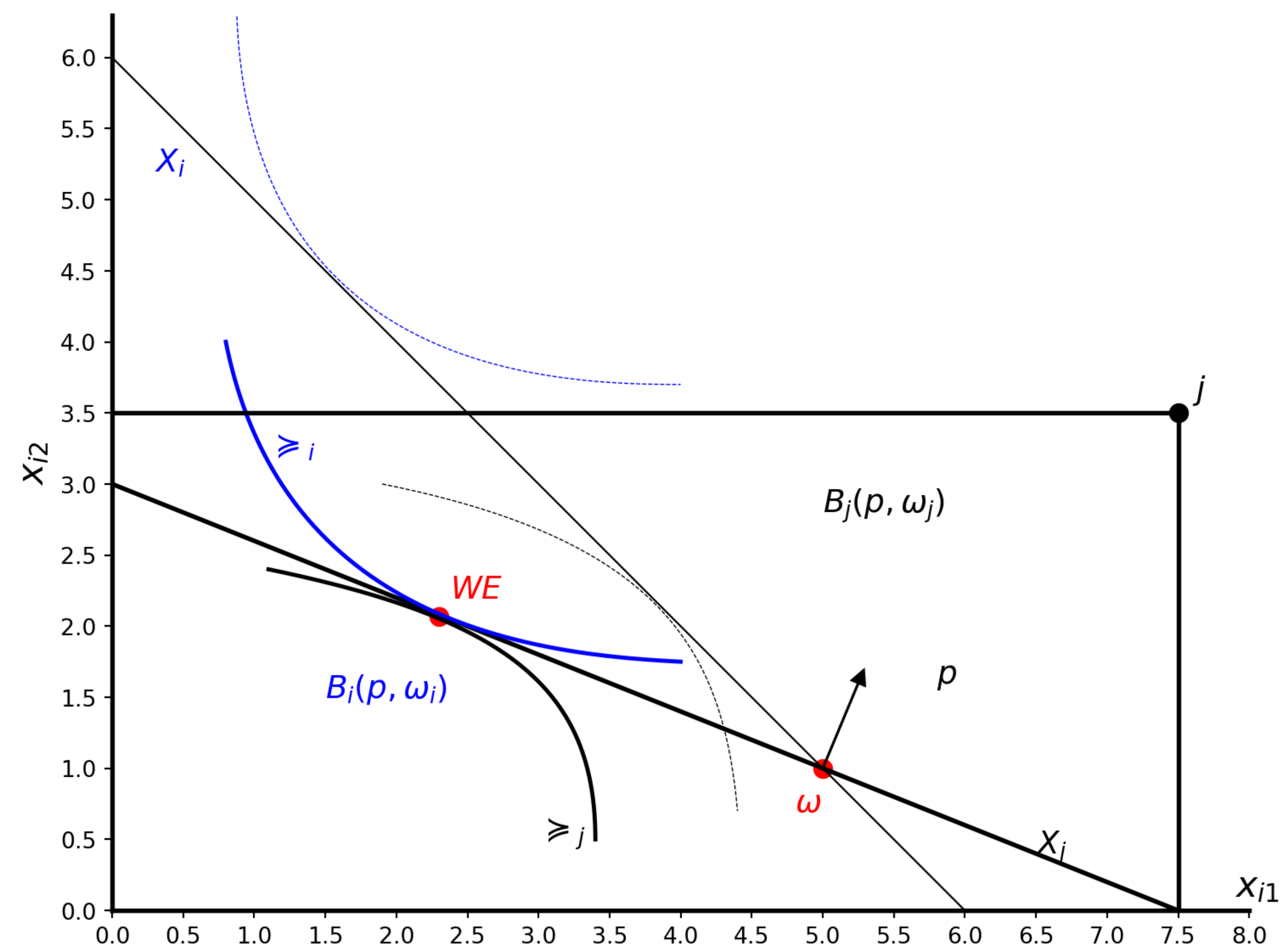
Edgeworth box, decentralization



Edgeworth box, i’s and j’s problem



Edgeworth box, WE



Robinson’s economy

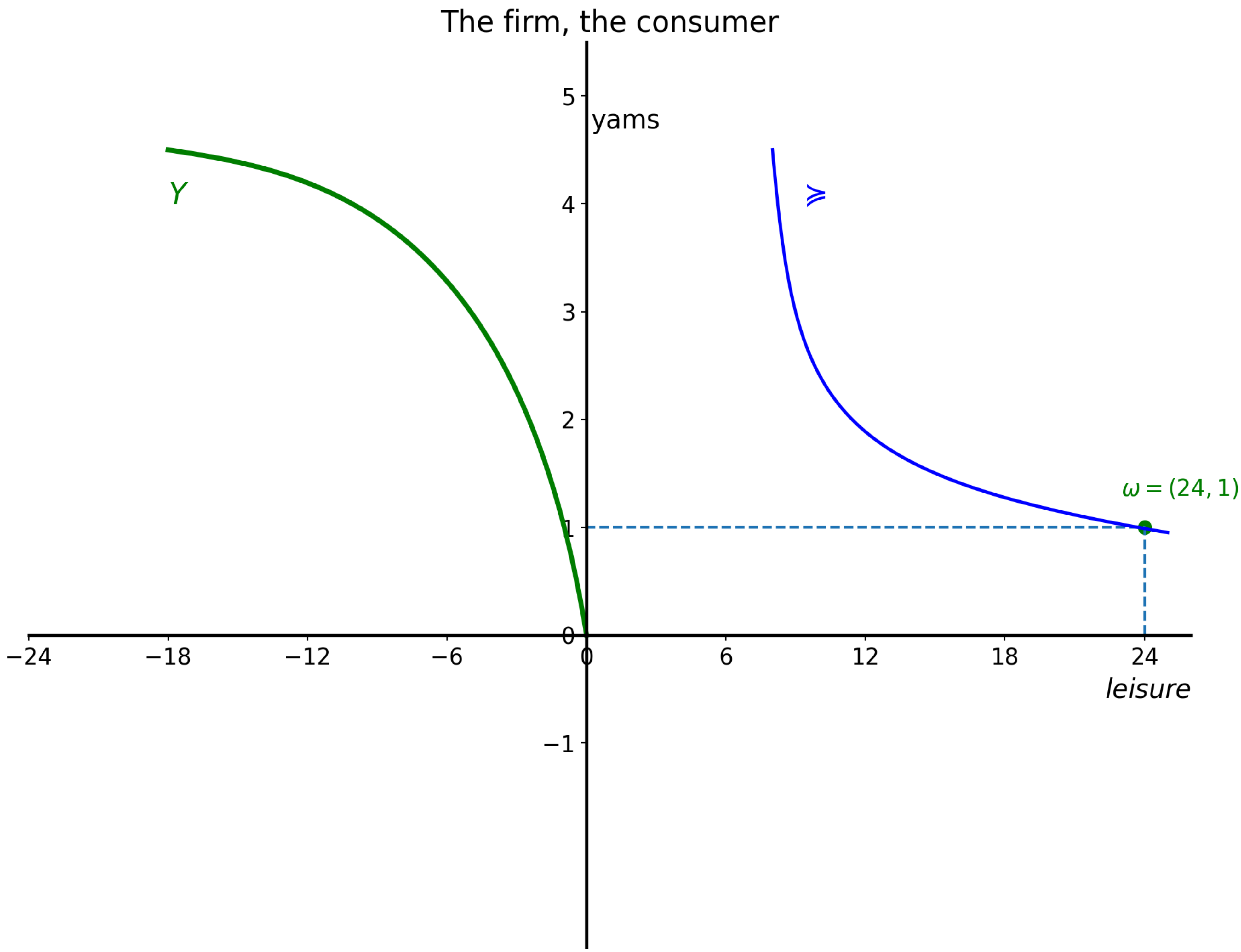
Robinson's economy

- Graphical representation of production economies.
- One firm, one agent (Robinson), and two goods, leisure x_1 and yams x_2.
- Robinson's consumption set:
  $X_R = \{(x_1, x_2) : (\forall i)\ x_i \geq 0, x_1 \leq 24\}$.
- Robinson's endowment: $\omega = (24, \omega_2)$.
- Preferences are complete, transitive, continuous, strongly monotone, strictly convex, (and as differentiable as convenient).

Robinson's firm, assumptions

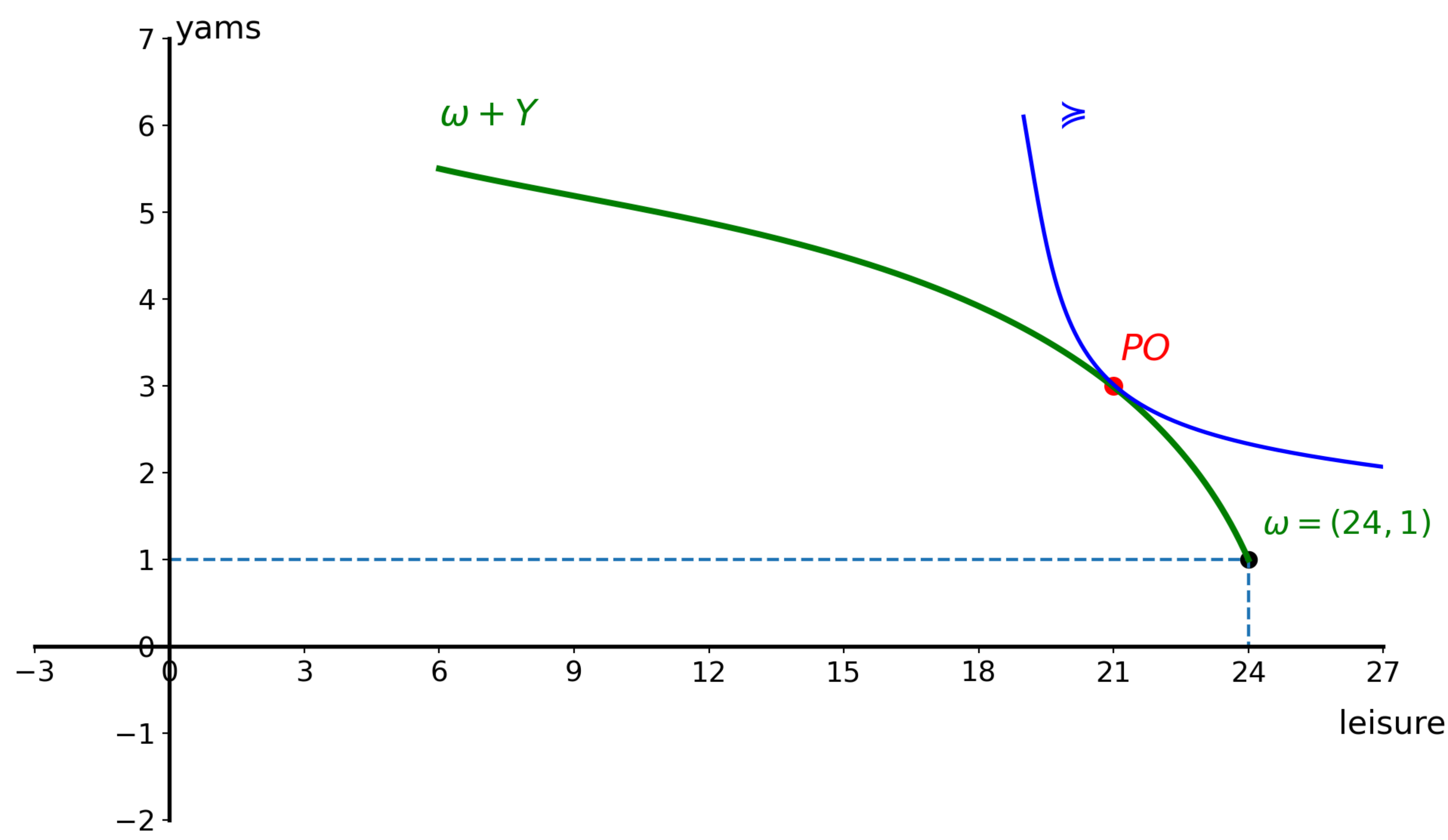
- The single firm has technology Y.
- Robinson is the sole owner of the firm.
- $0 \in Y$. Inactivity (i.e. no production) is possible.
- $Y + \mathbb{R}^L_- \subset Y$. Free disposal.
- $\mathbb{R}^L_+ \cap Y = \{0\}$. No free production.
- Y is convex. DRS or CRS.
- $y \in Y, y \neq 0 \implies -y \notin Y$. Irreversibility.
- And sufficiently differentiable when convenient.

Robinson’s economy, building the graph

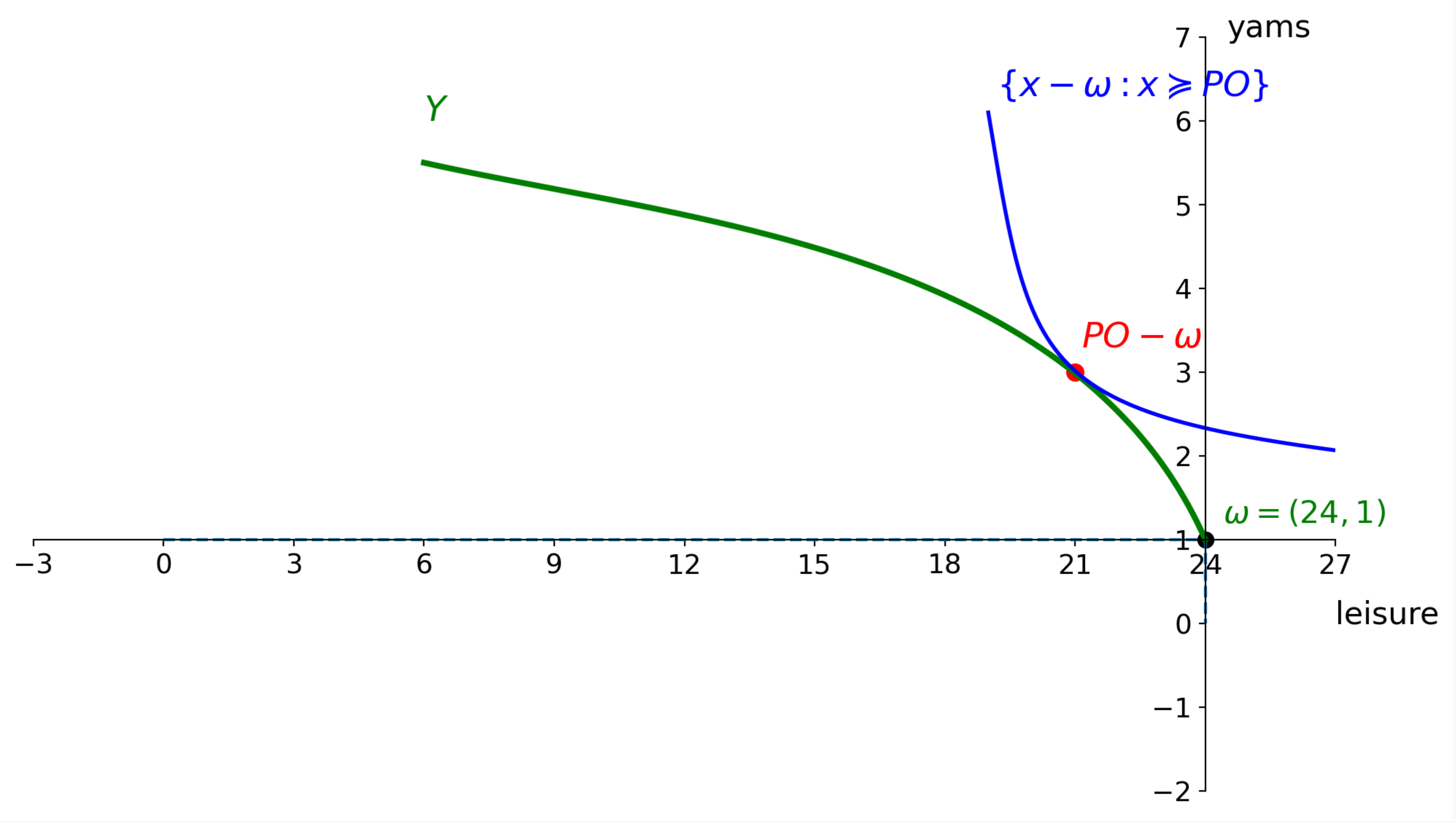


The firm, the consumer
yams
leisure
Y
≽
ω = (24, 1)
−24
−18
−12
−6
0
6
12
18
24
5
4
3
2
1
0
−1

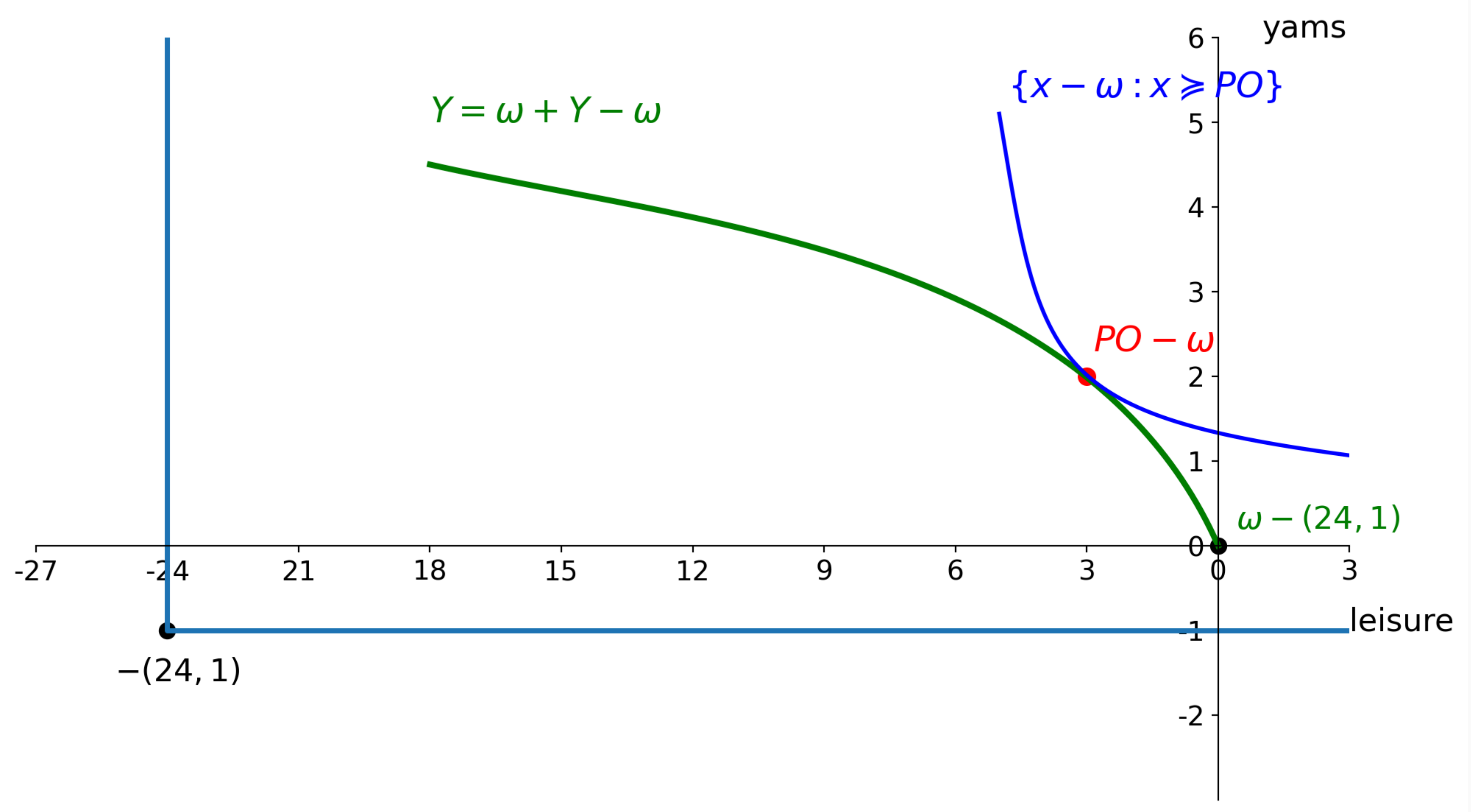
Robinson, planner's problem 1



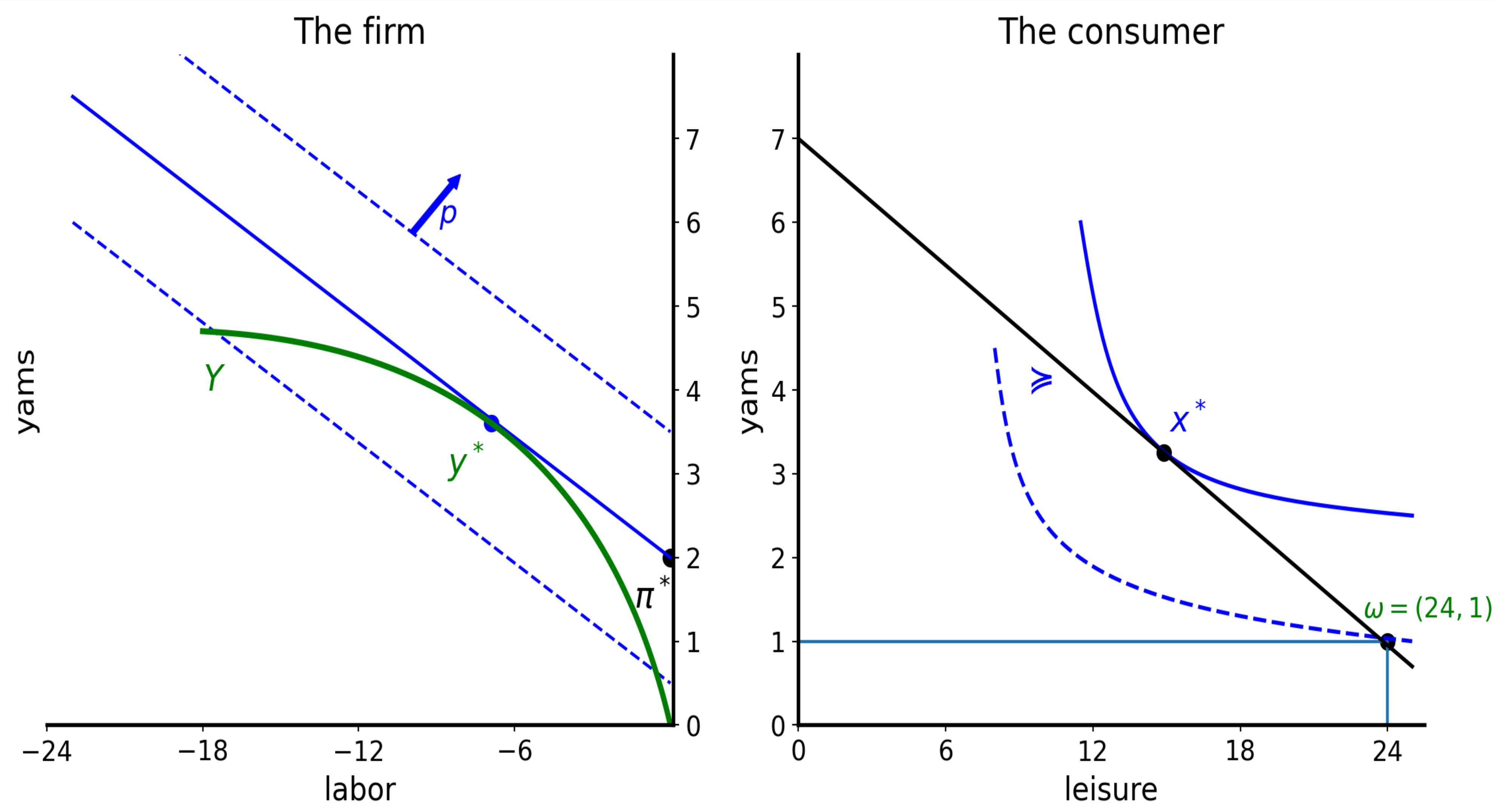
Robinson, planner’s problem 2



Robinson, planner’s problem 3



Decentralized problem



The firm's problem

$$\pi(p) = \max_{y \in Y} p \cdot y = \max_{y \in Y} p_2 y_2 + p_1 y_1 = \max_{y \in Y} p_2 y_2 - p_1 |y_1|$$

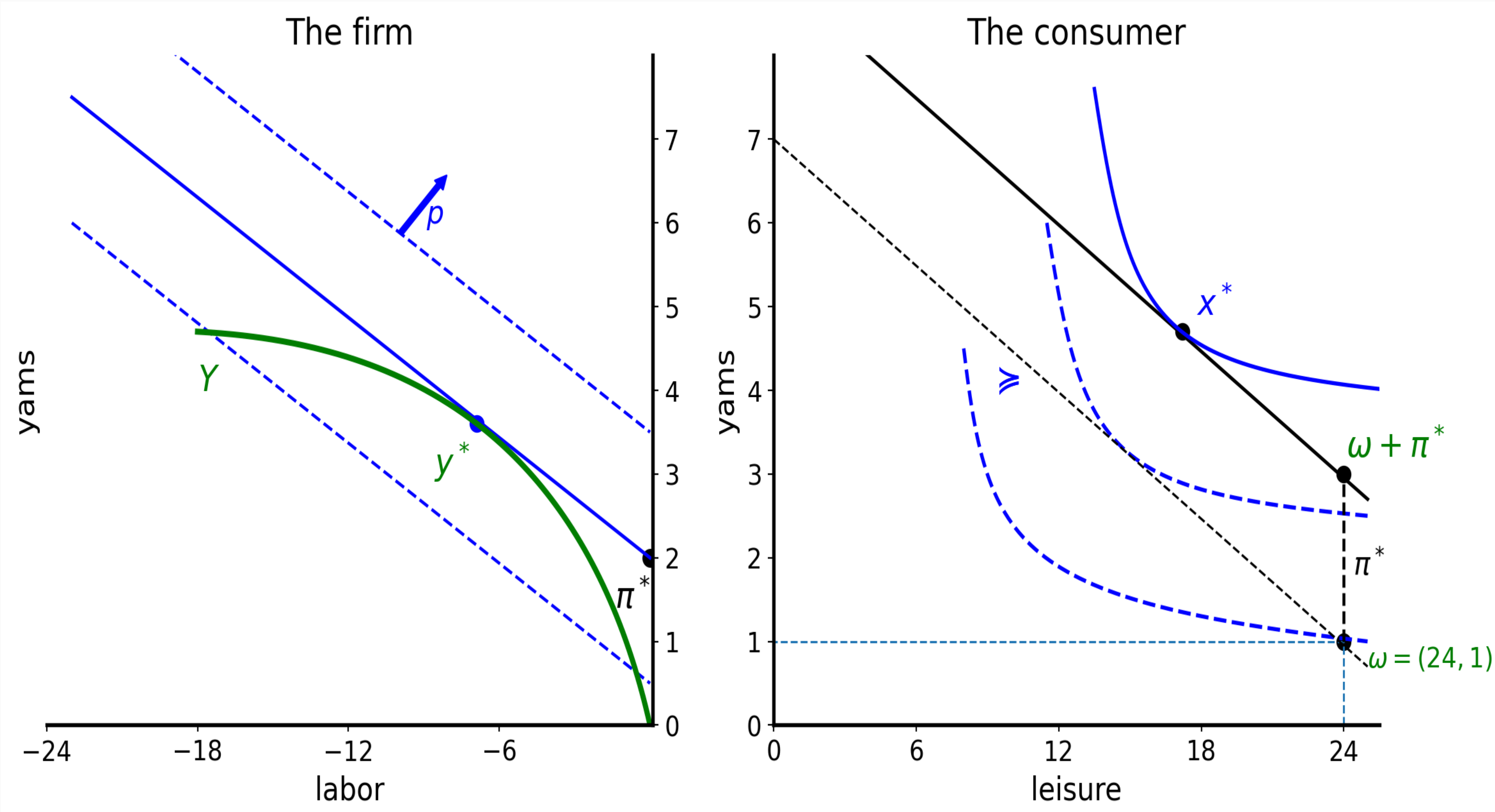
The isoprofit line corresponding to $\pi(p)$ is

$$\{y \in \mathbb{R}^2 : \pi(p) = p \cdot y\} \text{ or equivalently}$$
$$\left\{(y_1, y_2) \in \mathbb{R}^2 : y_2 = \frac{\pi(p)}{p_2} + \frac{p_1}{p_2}|y_1|\right\}$$

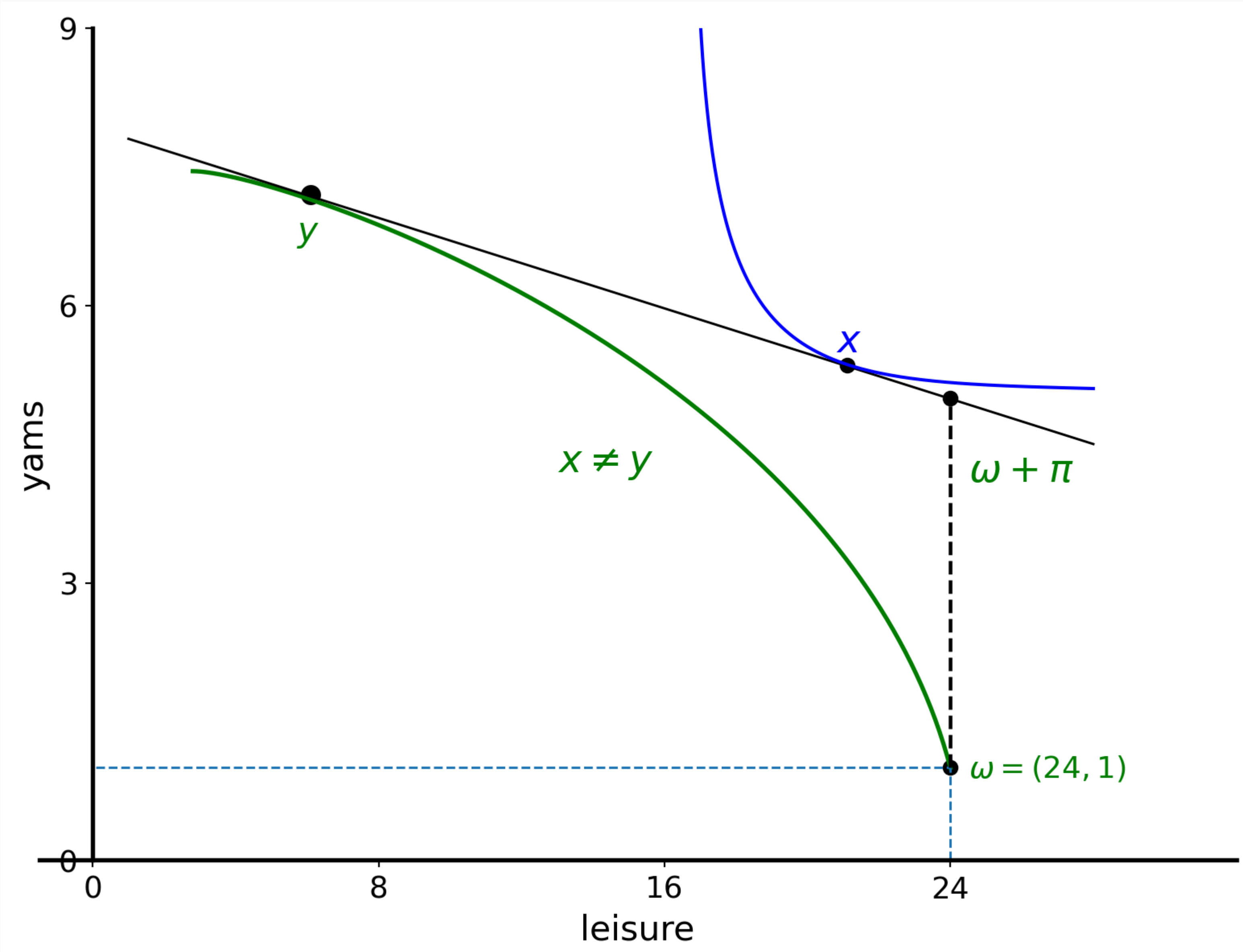
$$y_2^* = \frac{\pi(p)}{p_2} + \frac{p_1}{p_2}|y_1^*| = \pi(p) + p_1 |y_1^*| = \pi(p) = \pi^*$$

normalizing $p_2 = 1$ and choosing $y_1 = 0$ in the profit line.

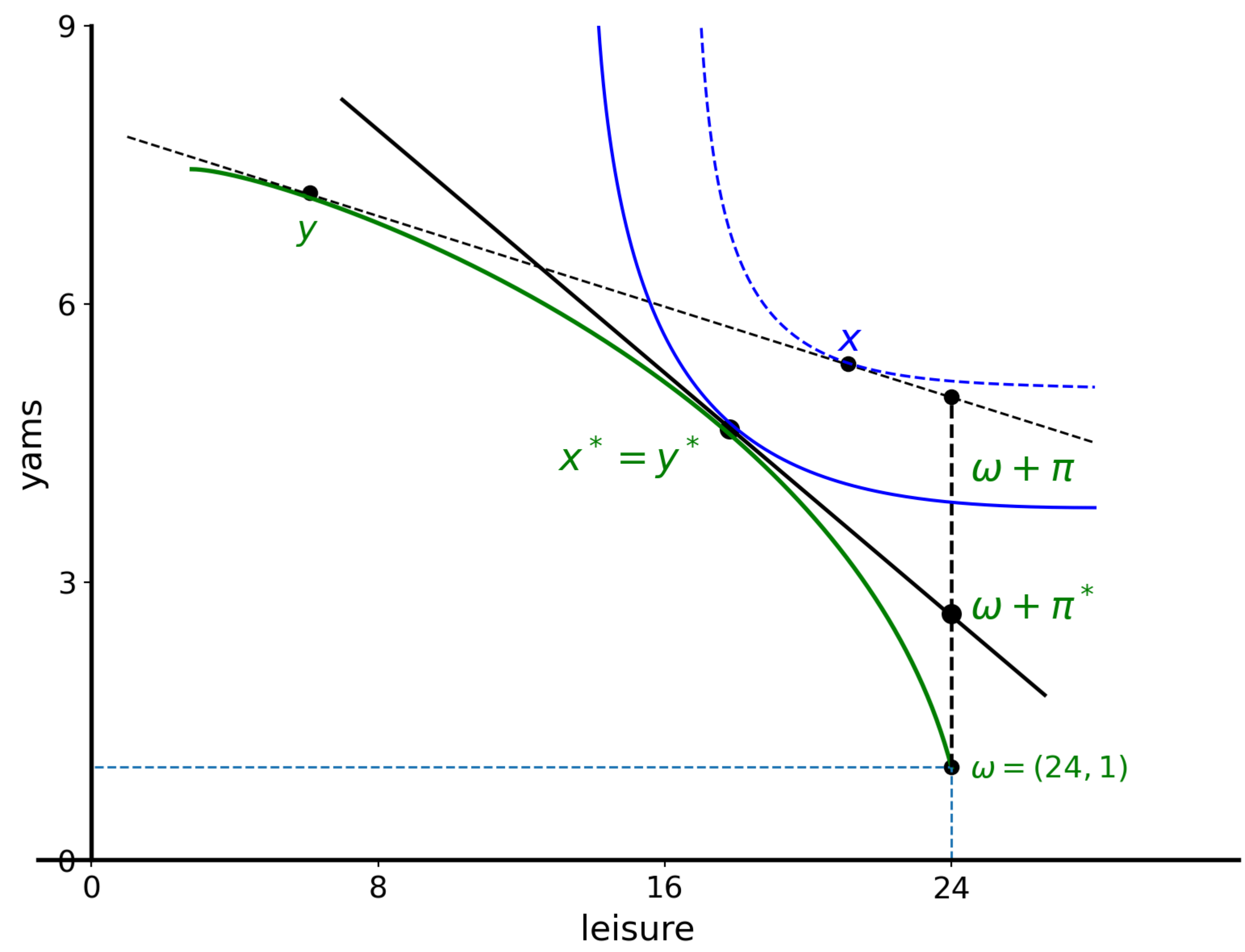
Decentralized problem and WE



Not a WE



A WE



Remarks

- The illustration of the welfare theorems also provides a method to identify WE; Negishi approach, (Negishi 1960).
- If Y is CRS in Robinson's economy, equilibrium prices are solely determined by technology provided the good is produced in equilibrium. CRS firms must earn zero profit.

Preferences do not affect prices or distribution of income.

References

Mas-Colell, Andreu, Michael Whinston, and Jerry Green. 1995. *Microeconomic Theory*. New York, Oxford: Oxford University Press.

Negishi, T. 1960. “Welfare Economics and the Existence of Equilibrium for a Competitive Economy.” *Metroeconomica* 12: 92–97.